// ZPWR-MIDI-FX //

# zpwr-midi-fx

v0.2.8

## MODULE REFERENCE

wire your own MIDI processor from note-stream blocks

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MenkeTechnologies

# Contents

# Overview

Where a fixed chord + arp tool is a single black box, zpwr-midi-fx is a patchable grid of MIDI modules — the same modular-patch model as zpwr-fx, but the signal flowing between blocks is a stream of note events, not audio. Wire MIDI In → blocks → Out, cross-modulate block parameters from a mod matrix, and drive it all from soft keys, LFOs, envelopes and live MIDI / MPE expression.

It is built on the shared zpwr-patch-core (zpc) graph — the same engine as zpwr-fx and zpwr-synth, instantiated on a note-event stream instead of audio samples. Part of the MenkeTechnologies audio stack alongside zpwr-synth and zpwr-fx.

## Formats & Platforms

VST3 AU (macOS) CLAP (note-effect) Standalone

Built on JUCE 8.0.13 with a cyberpunk WebView UI. On macOS the build is a universal binary (x86_64;arm64), so the VST3 / AU / CLAP load on both Intel and Apple Silicon hosts. On Windows it ships VST3 + CLAP (AU is macOS-only); x64 and ARM64 are separate builds. The audio path is pass-through, so it drops in front of any instrument.

# Patch Model

A dynamic set of blocks (add / remove / reorder), each with a module type, note-stream inputs plus a scalar `Mod` input (unbounded summed cables per slot), up to six parameters, and one output. Inputs select any source; the two outputs (`Out A`, `Out B`) merge to the plugin's MIDI out. The graph is evaluated once per audio block in topological order, and note-offs are scheduled across block boundaries so nothing ever hangs.

MIDI In □ [B1 Chord] □ [B2 Arp] □ Out A □ MIDI Out

- ⚡ EZ WIRE auto-wires MIDI In → your blocks → Out A so a beginner gets a working chain without touching cables.
- INIT unplugs every cable & mod while keeping the blocks; 🗑️ blanks the whole patch.
- INPUTS column exposes every patchable source as a jack — the note input, all scalar modulators (Random + performance / MPE controllers) and the soft keys — so anything usable in the mod matrix can be cabled directly into a block.
- Cables are drawn between jacks and can be dragged to re-patch. Right-click a cable for its editor: a Level control (scales note velocity on that connection — `0` mutes it, and cable brightness / width follow the level), a Colour swatch, and Disconnect.

Every control has a hover tooltip explaining what it does. The patch (blocks, routing, mods) is zpc JSON the WebView edits and the plugin persists in its state.

# Modules

111 note-stream modules across 10 categories — harmony, sequencing, probability, routing, control sources, dynamics, expression, tuning, MPE / voicing, plus a family of cellular-automaton sequencers (Game of Life, Brian's Brain, Langton's Ant). The reference lists every block with its inputs and parameters, generated from the live registry so it never drifts.

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- Probability 1
- Routing 9
- Sequencing 35
- Tuning 1
- Voicing 5

Representative blocks: Chord (one key → a voiced chord, 166 types, with inversion / spread / octave-double / transpose / strum), Arp (host-synced arpeggiator, 9 play modes, divisions, gate, octaves, swing, latch), Scale (snap notes to the nearest in-key pitch), SeqEuclid (Euclidean rhythm gate via Bjorklund), Chance (per-note probability gate), Harmonize (stack fixed intervals), Echo (MIDI delay with feedback), MidiLFO / MidiEnv (scalar control sources), Humanize, Strum, MPEConvert and the cellular-automaton sequencers MidiGameOfLife, MidiBrianBrain, MidiLangton.

# Modulation

A dynamic list of routes, each mapping a scalar source → any block parameter with a depth. Routing rebuilds the graph; depth is a live, lock-free tweak. The same sources are available on each block's Mod input. Sources:

- Soft Keys — an expandable pool of host-automatable macros (ceiling 32; 16 active by default; + / − to add / remove, active count saved in plugin state).
- Auto-param pool — a reserved pool of 96 Auto N host params; right-click any block param to bind it (the Kontakt/Reaktor model), so any module parameter becomes host-automatable / CC-mappable. 32 soft keys + 96 auto params = 128 CC-mappable slots, one per MIDI CC.
- LFO and Env block outputs, and Random.
- Performance controllers — Mod Wheel, Pitch Bend, Aftertouch, Velocity, Expression (CC11), Sustain (CC64).
- MPE — per-note Bend, Pressure and Slide (CC74), read from the most recently active note's channel.

## Soft Keys & Automation

The host-automatable parameters are the soft keys — an expandable pool of macros (ceiling 32; 16 active by default, the active count saved in plugin state), with + / − to add / remove — plus a reserved pool of 96 Auto N host params: right-click any block param to bind it (the Kontakt/Reaktor model), so any module parameter becomes host-automatable / CC-mappable. 32 soft keys + 96 auto params = 128 CC-mappable slots, one per MIDI CC. Everything else lives in the patch, saved as zpc JSON in plugin state. Modulation is the patching itself: any LFO / Env block output or soft key patched into a block input — or mod-routed to a param — is a mod-matrix connection.

# User Interface

The editor is a WebView hosting the shared zpwr-patch-core cyberpunk patcher — the same UI as zpwr-fx and zpwr-synth: drag-to-patch neon cables (fan-out, feedback, per-cable level + colour), a dynamic block grid (add / delete any number, no fixed block count), and a per-param mod matrix. Every control has a hover tooltip explaining what it does. Tabs:

- PATCH — the block grid + cable routing + an expandable row of soft-key macro knobs. The toolbar carries ⚡ EZ WIRE (auto-wire MIDI In → your blocks → Out A), INIT (unplug every cable & mod, keep the blocks) and 🗑️ (blank the whole patch). The INPUTS column exposes every patchable source as a jack — the note input, all scalar modulators (Random + performance / MPE controllers) and the soft keys. Cables are drawn between jacks and can be dragged to re-patch; right-click a cable for its editor (Level, Colour, Disconnect).
- PERFORM — a macros-and-pads play surface with no patching (see Perform).
- PRESETS — browse and manage the factory bank and user patches (see Presets).

## Perform

The PERFORM tab is a macros-and-pads view with no patching, for live play:

- Preset Morph — a 4-corner XY pad (A/B/C/D) that bilinearly interpolates between four captured presets; X/Y are reserved host params (morphX / morphY), so the morph is host-automatable.
- Orb — angle picks one of 8 random scenes (per-macro offset vectors), distance from centre scales intensity; record / loop the orb motion while you drag.
- XY macro pads — each pad drives a pair of soft keys; per-pad HOLD keeps the dot, SPRING snaps both axes back to centre on release.
- Macro knobs — a dial per active soft key; Snapshots — 8 slots capturing the whole macro surface (per-plugin localStorage).
- On-screen keyboard — global Key + Scale quantize of incoming notes and a Chord selector (Off / Oct / 5th / Maj / Min / Maj7 / Min7 / Sus4 / Power), plus a global arp (mode / rate / latch).

MIDI In response — PROGRAM and BANK toggles (both default ON): incoming Program Change switches presets, with Bank Select (CC0 MSB / CC32 LSB) captured for the next Program Change. Both are saved in plugin state.

## Presets

A factory bank ships with the plugin (Chord Arp, Strummed Chords, Euclidean Pulse, Fifth Harmonizer, Arp Echo, Random Walk, Step Melody, Tone Cluster, Jazz Voicing, MPE Spread and more) spanning arps, harmony, rhythm, generative and FX chains. User patches are saved as `zpwr-midi-fx/Presets/*.zmfxpatch` under the user application data directory and managed from the PRESETS tab.

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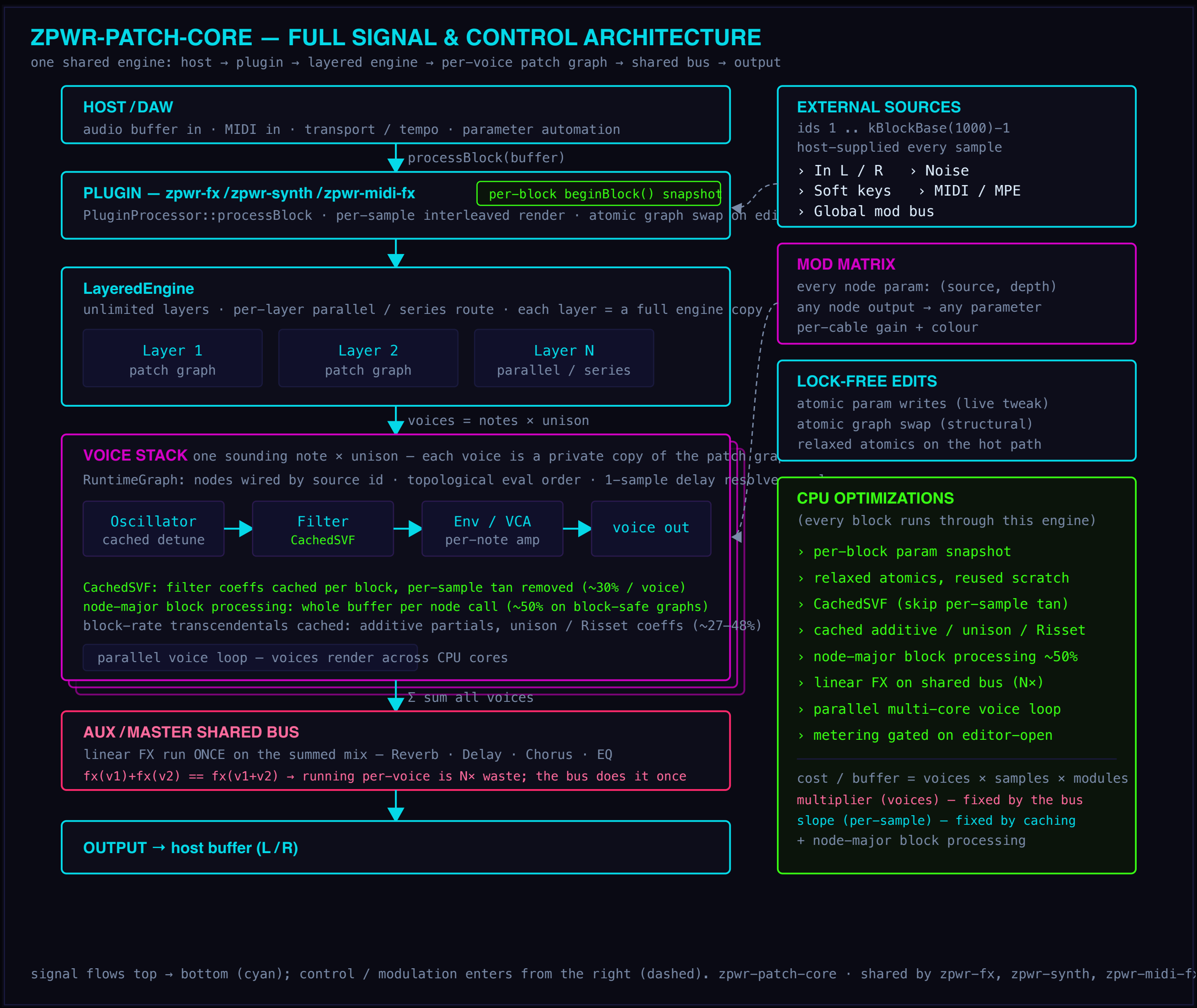
# Architecture

ZPWR-MIDI-FX runs on the shared zpwr-patch-core engine. The full signal and control path — from the host buffer, through the layered engine and the per-voice patch graph, onto the shared bus, and back out — is shown below.



## Signal Sources

External sources you can cable into any block input or modulation route.

1 MIDI In 3 Random Mod Wheel Pitch Bend Aftertouch Velocity Expression Sustain MPE Bend MPE Pressure MPE Slide

# Modulation

13 modules

## Generative 1 input

Probabilistic melodic random walk synced to a rate.

| Param | Range | Default | Unit |
|---|---|---|---|
| Rate | 1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T | | – |
| Range | 1 – 12 | 3 | st |
| Low | 0 – 127 | 48 | – |
| High | 0 – 127 | 72 | – |
| Probability | 0 – 1 | 0.7 | – |

## MidiLFO 1 input

Low-frequency modulation source (sine/tri/saw/square) on the scalar control output.

| Param | Range | Default | Unit |
|---|---|---|---|
| Rate | 0.01 – 20 | 1 | Hz |
| Shape | Sine · Triangle · Saw · Square | | – |
| Depth | 0 – 1 | 1 | – |

## MidiEnv 1 input

ADSR envelope triggered by the note gate, on the scalar control output.

| Param | Range | Default | Unit |
|---|---|---|---|
| Attack | 1 – 5000 | 5 | ms |
| Decay | 1 – 5000 | 200 | ms |
| Sustain | 0 – 1 | 0.7 | – |
| Release | 1 – 5000 | 300 | ms |

## MidiRandom 1 input

Generates random notes within a pitch range at a synced rate.

| Param | Range | Default | Unit |
|---|---|---|---|
| Rate | 1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T | | – |
| Low Note | 0 – 127 | 48 | – |
| High Note | 0 – 127 | 72 | – |
| Gate | 0 – 2 | 0.5 | – |
| Velocity | 1 – 127 | 100 | – |

## MidiSampleHold 1 input

Samples and holds the incoming control value on each note.

### MidiSlew 1 input

Smooths (glides) the control value over a set time.

| Param | Range | Default | Unit |
|---|---|---|---|
| Time | 0 – 2000 | 100 | ms |

### RandOctave 1 input

Randomly shifts notes by up to +/- Range octaves with a chance.

| Param | Range | Default | Unit |
|---|---|---|---|
| Range | 1 – 3 | 1 | – |
| Chance | 0 – 1 | 1 | – |

### CCLFO 1 input

Continuous CC LFO generator (sine/triangle/saw/square) emitted on a chosen CC at a free Rate - automates a synth parameter over MIDI, where MidiLFO drives only the internal mod-matrix scalar.

| Param | Range | Default | Unit |
|---|---|---|---|
| CC | 0 – 127 | 1 | – |
| Rate | 0.01 – 20 | 1 | Hz |
| Shape | Sine · Triangle · Saw · Square | | – |
| Depth | 0 – 1 | 1 | – |

### CCSeq 1 input

Tempo-synced step CC sequencer: one CC value per synced step cycling an 8-step shape (overridable via the node's config list) - a modulation sequencer in the CC domain, where StepSeqMidi sequences notes.

| Param | Range | Default | Unit |
|---|---|---|---|
| CC | 0 – 127 | 1 | – |
| Rate | 1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T | | – |

### VelToCC 1 input

Turns each note's velocity into a CC value emitted just before the note, so a synth's filter or level opens with how hard you played - dynamics routed to expression, where Note2CC sends pitch.

| Param | Range | Default | Unit |
|---|---|---|---|
| CC | 0 – 127 | 11 | – |

### CCGlide 1 input

Portamento for a CC: smooths the watched CC toward each newly received value over Time, emitting interpolated CC each block - de-zippers a stepped controller, where MidiSlew only glides the internal scalar.

| Param | Range | Default | Unit |
|---|---|---|---|
| CC | 0 – 127 | 1 | – |
| Time | 1 – 2000 | 120 | ms |

## PressSwell 1 input

Generates a channel-pressure (aftertouch) swell that rises to Max over Rise while any note is held, snapping back to zero on release - an automatic crescendo for pads and strings, where ATProc only transforms incoming aftertouch.

| Param | Range | Default | Unit |
|---|---|---|---|
| Rise | 1 - 5000 | 800 | ms |
| Max | 0 - 1 | 1 | – |

## PitchDrift 1 input

Nudges each note's pitch by a small random detune up to +/-Cents (per-note, polyphonic) - the gentle oscillator instability of analog gear, applied to pitch where Humanize only scatters timing and velocity.

| Param | Range | Default | Unit |
|---|---|---|---|
| Cents | 0 - 50 | 8 | c |

# Dynamics

10 modules

## Velocity 1 input

Scales and offsets note velocity, with a modulation amount.

| Param | Range | Default | Unit |
|---|---|---|---|
| Scale | 0 – 2 | 1 | – |
| Offset | -64 – 64 | 0 | – |
| Mod Amt | -64 – 64 | 0 | – |

## VelCurve 1 input

Reshapes velocity through a power curve.

| Param | Range | Default | Unit |
|---|---|---|---|
| Curve | 0.2 – 5 | 1 | – |

## Humanize 1 input

Adds random timing and velocity jitter for a human feel.

| Param | Range | Default | Unit |
|---|---|---|---|
| Timing | 0 – 50 | 8 | ms |
| Velocity | 0 – 1 | 0.3 | – |

## Accent 1 input

Boosts velocity on every Nth note.

| Param | Range | Default | Unit |
|---|---|---|---|
| Every | 1 – 16 | 4 | – |
| Amount | 0 – 1 | 0.5 | – |

## VelClip 1 input

Clamps velocity between Min and Max.

| Param | Range | Default | Unit |
|---|---|---|---|
| Min | 1 – 127 | 1 | – |
| Max | 1 – 127 | 127 | – |

## Ramp 1 input

Velocity ramp over a run of notes, ascending or descending.

| Param | Range | Default | Unit |
|---|---|---|---|
| Length | 2 - 16 | 8 | – |
| Depth | 0 - 1 | 0.6 | – |
| Direction | Up · Down | | – |

## VelCompress 1 input

Velocity compressor: velocities above Threshold are pulled toward it by Ratio (1 = off, 8 = hard), narrowing the dynamic range smoothly instead of hard-clamping like VelClip.

| Param | Range | Default | Unit |
|---|---|---|---|
| Threshold | 1 - 127 | 64 | – |
| Ratio | 1 - 8 | 2 | – |

## VelInvert 1 input

Flips the velocity scale around its midpoint (soft <-> loud) by Amount, so your accents become ghost notes and the ghosts become accents - a velocity mirror, blendable to taste.

| Param | Range | Default | Unit |
|---|---|---|---|
| Amount | 0 - 1 | 1 | – |

## VelKey 1 input

Keyboard velocity tracking: scales each note's velocity by how far its pitch sits from a Center note, Tilt setting how strongly and which way - higher notes louder (or softer), the dynamic key-tracking of real instruments.

| Param | Range | Default | Unit |
|---|---|---|---|
| Center | 0 - 127 | 60 | – |
| Tilt | -1 - 1 | 0.5 | – |

## VelLength 1 input

Note duration scales with velocity: hard notes ring for Max ms, soft notes for Min (set Max < Min to make accents staccato) - the played articulation where dynamics shape note length.

| Param | Range | Default | Unit |
|---|---|---|---|
| Min | 1 - 2000 | 60 | ms |
| Max | 1 - 2000 | 500 | ms |

# Control

5 modules

## CCMap 1 input

Remaps a CC number and scales/offsets its value.

| Param | Range | Default | Unit |
|---|---|---|---|
| From | -1 – 127 | 1 | – |
| To | 0 – 127 | 11 | – |
| Scale | 0 – 2 | 1 | – |
| Offset | -1 – 1 | 0 | – |

## Note2CC 1 input

Emits a CC from each note (pitch or velocity).

| Param | Range | Default | Unit |
|---|---|---|---|
| CC | 0 – 127 | 1 | – |
| Source | Pitch · Velocity | | – |

## CC2Note 1 input

A watched CC crossing a threshold triggers a note.

| Param | Range | Default | Unit |
|---|---|---|---|
| Watch | 0 – 127 | 1 | – |
| Note | 0 – 127 | 60 | – |
| Threshold | 0 – 1 | 0.5 | – |

## BendProc 1 input

Scales and/or inverts incoming pitch-bend.

| Param | Range | Default | Unit |
|---|---|---|---|
| Scale | 0 – 2 | 1 | – |
| Invert | 0 – 1 | 0 | – |

## ATProc 1 input

Scales aftertouch, optionally converting it to a CC.

| Param | Range | Default | Unit |
|---|---|---|---|
| Scale | 0 – 2 | 1 | – |
| To CC | -1 – 127 | -1 | – |

# Expression

3 modules

## Scoop 1 input

Bends up into each struck note from Depth below, resolving to pitch over Time - the vocal/brass scoop, a pitch-bend ramp that ends at centre so no bend is left hanging.

| Param | Range | Default | Unit |
|---|---|---|---|
| Depth | 0 – 1 | 0.3 | – |
| Time | 1 – 1000 | 80 | ms |

## Fall 1 input

Jazz fall-off: on release the pitch sweeps down by Depth over Time while the note rings, then the note releases and the bend snaps back to centre. The note-off is delayed to the end of the fall.

| Param | Range | Default | Unit |
|---|---|---|---|
| Depth | 0 – 1 | 0.5 | – |
| Time | 10 – 2000 | 300 | ms |

## MidiVibrato 1 input

A pitch-bend LFO that fades in after an Onset delay while any note is held - the delayed, played-in vibrato of a string or voice rather than an instant wobble. Rate in Hz, Depth as bend amount.

| Param | Range | Default | Unit |
|---|---|---|---|
| Rate | 0.1 – 12 | 5 | Hz |
| Depth | 0 – 1 | 0.15 | – |
| Onset | 0 – 2000 | 300 | ms |

# Harmony

21 modules

## Chord 1 input

Turns each incoming note into a chord (165 types) with inversion, spread, octave doubling and strum.

| Param | Range | Default |
|---|---|---|
| Type | Unison · Octave · 5th (Power) · 5th + Oct · Tritone · Major · Minor · Diminished · Augmented · Major (1st inv) · Major ( | |
| Inversion | 0 – 4 | 0 |
| Spread | 0 – 2 | 0 |
| Octave x2 | 0 – 2 | 0 |
| Transpose | -24 – 24 | 0 |
| Strum | 0 – 200 | 0 |

## Scale 1 input

Quantizes incoming notes to the nearest degree of the chosen root and scale.

| Param | Range | Default |
|---|---|---|
| Root | C · C# · D · D# · E · F · F# · G · G# · A · A# · B | |
| Scale | Chromatic · Major · Natural Minor · Harmonic Minor · Melodic Minor · Dorian · Phrygian · Lydian · Mixolydian · Locrian · Pen | |

## Transpose 1 input

Shifts every note by a fixed semitone amount plus a modulatable offset.

| Param | Range | Default | Unit |
|---|---|---|---|
| Semitones | -24 – 24 | 0 | st |
| Mod Amt | -24 – 24 | 0 | st |

## Harmonize 1 input

Adds up to three fixed-interval harmony voices to each note.

| Param | Range | Default | Unit |
|---|---|---|---|
| Interval 1 | -24 – 24 | 7 | st |
| Interval 2 | -24 – 24 | 0 | st |
| Interval 3 | -24 – 24 | 0 | st |

## Drone 1 input

Doubles every note with a fixed pedal root.

| Param | Range | Default | Unit |
|---|---|---|---|
| Root | 0 – 127 | 36 | – |

## MidiCluster 1 input

Adds symmetric tone-cluster voices above and below each note.

| Param | Range | Default | Unit |
|---|---|---|---|
| Width | 0 - 4 | 1 | – |
| Spread | 1 - 4 | 1 | st |

## Glissando 1 input

Fills a stepwise run between consecutive notes.

| Param | Range | Default | Unit |
|---|---|---|---|
| Rate | 5 - 200 | 40 | ms |

## ChordMem 1 input

Each key plays a stored chord shape (semitone intervals from node config).

## ChordProg 1 input

Each note triggers the next chord in a stored root progression (node config).

## ChordVoicing 1 input

Re-voices each note into a jazz shape (open / shell / quartal / spread).

| Param | Range | Default | Unit |
|---|---|---|---|
| Voicing | Open · Shell · Quartal · Spread | | – |

## Octave 1 input

Doubles each note up and/or down by whole octaves.

| Param | Range | Default | Unit |
|---|---|---|---|
| Up | 0 - 3 | 1 | – |
| Down | 0 - 3 | 0 | – |

## MidiFold 1 input

Folds out-of-range notes back inside a Low..High window.

| Param | Range | Default | Unit |
|---|---|---|---|
| Low Note | 0 - 127 | 48 | – |
| High Note | 0 - 127 | 72 | – |

## Invert 1 input

Mirrors pitch around a pivot note.

| Param | Range | Default | Unit |
|---|---|---|---|
| Pivot | 0 – 127 | 60 | – |

## Diatonic 1 input

Transposes by scale DEGREES within Root/Scale (a third, a fifth...) instead of by raw semitones like Transpose - the shift always lands back in key. Mod Amt adds a modulatable degree offset.

| Param | Range | Default |
|---|---|---|
| Root | C · C# · D · D# · E · F · F# · G · G# · A · A# · B | |
| Scale | Chromatic · Major · Natural Minor · Harmonic Minor · Melodic Minor · Dorian · Phrygian · Lydian · Mixolydian · Locrian · P | |
| Degrees | -7 – 7 | 2 |
| Mod Amt | -7 – 7 | 0 |

## DiatonicChord 1 input

Auto-harmoniser: builds the in-key chord on every note by stacking diatonic thirds (triad, 7th...) along Root/Scale, so single-finger melodies become correct chords in the key.

| Param | Range | Default |
|---|---|---|
| Root | C · C# · D · D# · E · F · F# · G · G# · A · A# · B | |
| Scale | Chromatic · Major · Natural Minor · Harmonic Minor · Melodic Minor · Dorian · Phrygian · Lydian · Mixolydian · Locrian · Pen | |
| Notes | 2 – 4 | 3 |

## NeoRiemann 1 input

Neo-Riemannian triad transforms: builds a major/minor triad on each note and applies P (parallel), L (leading-tone) or R (relative) - the maximally-smooth chord moves of late-Romantic and film harmony, each sharing two common tones with the input.

| Param | Range | Default | Unit |
|---|---|---|---|
| Quality | Major · Minor | | – |
| Transform | None · P · L · R | | – |

## Counterpoint 1 input

Adds a second voice in contrary motion - when the melody rises the counter voice falls and vice-versa - quantised to Root/Scale. Interval sets the starting separation.

| Param | Range | Default |
|---|---|---|
| Root | C · C# · D · D# · E · F · F# · G · G# · A · A# · B | |
| Scale | Chromatic · Major · Natural Minor · Harmonic Minor · Melodic Minor · Dorian · Phrygian · Lydian · Mixolydian · Locrian · | |
| Interval | 1 – 24 | 7 |

## Bassline 1 input

Follows the chord with a mono bass voice: the lowest held note dropped Octaves down, retriggered whenever the lowest note changes. Passes the original notes through and adds the bass beneath them.

| Param | Range | Default | Unit |
|---|---|---|---|
| Octaves | 1 – 3 | 1 | – |

| Param | Range | Default | Unit |
| --- | --- | --- | --- |
| Velocity | 1 – 127 | 100 | – |

## VoiceLead 1 input

Smooth voice leading: places each note's pitch class in the octave nearest the previous output, so a line or chord progression moves by the smallest interval. Range caps the leap; past it the original octave is kept.

| Param | Range | Default | Unit |
| --- | --- | --- | --- |
| Range | 1 – 12 | 7 | st |

## AutoInvert 1 input

Cycles the inversion of each struck chord: every trigger rotates which of the lowest notes jump an octave (by Rotate), so a repeated chord keeps climbing through its inversions - movement ChordVoicing's static shapes don't give.

| Param | Range | Default | Unit |
| --- | --- | --- | --- |
| Rotate | 1 – 4 | 1 | – |

## ChordLock 1 input

Quantizes every note to the nearest tone of a fixed chord (Root + Type), tighter than Scale's scale-quantize - locks an improvisation to a chord's notes.

| Param | Range | Default |
| --- | --- | --- |
| Root | C · C# · D · D# · E · F · F# · G · G# · A · A# · B | |
| Type | Unison · Octave · 5th (Power) · 5th + Oct · Tritone · Major · Minor · Diminished · Augmented · Major (1st inv) · Major (2nd | |

# Probability

1 modules

## Chance 1 input

Probabilistic gate: lets each note through with the set probability.

| Param | Range | Default | Unit |
|---|---|---|---|
| Probability | 0 – 1 | 0.8 | – |
| Mod Amt | -1 – 1 | 0 | – |

# Routing

11 modules

## Split 1 input

Keyboard split: notes below the split point transpose by Low, the rest by High.

| Param | Range | Default | Unit |
|---|---|---|---|
| Split | 0 – 127 | 60 | – |
| Low | -24 – 24 | 0 | st |
| High | -24 – 24 | 0 | st |

## Transformer 1 input

Rule: notes within a range are transposed and velocity-scaled.

| Param | Range | Default | Unit |
|---|---|---|---|
| Low | 0 – 127 | 0 | – |
| High | 0 – 127 | 127 | – |
| Transpose | -24 – 24 | 0 | st |
| VelScale | 0 – 2 | 1 | – |

## Merge 2 inputs

Passes both note-stream inputs through, merged into one stream.

## NoteFilter 1 input

Passes only notes inside a key and velocity window.

| Param | Range | Default | Unit |
|---|---|---|---|
| Low Note | 0 – 127 | 0 | – |
| High Note | 0 – 127 | 127 | – |
| Low Vel | 1 – 127 | 1 | – |
| High Vel | 1 – 127 | 127 | – |

## MidiLatch 1 input

Holds notes after release until the next note arrives.

## Channel 1 input

Reassigns notes to a fixed MIDI channel.

| Param | Range | Default | Unit |
|---|---|---|---|
| Channel | 1 – 16 | 1 | – |

## FixedNote 1 input

Forces every note to a single fixed pitch (drum / trigger).

| Param | Range | Default | Unit |
|---|---|---|---|
| Note | 0 – 127 | 36 | – |

## KeySwitch 1 input

Splits the keyboard at a note; one zone passes, the other is filtered.

| Param | Range | Default | Unit |
|---|---|---|---|
| Split | 0 – 127 | 60 | – |
| Low Zone | High passes · Low passes | | – |

## NoteLimit 1 input

Polyphony cap: never lets more than Max notes sound at once, stealing the oldest (FIFO note-off) when a new note would exceed the limit - tames dense chords and runaway generators.

| Param | Range | Default | Unit |
|---|---|---|---|
| Max | 1 – 16 | 4 | – |

## PolyChan 1 input

Round-robins each new note to the next of Voices MIDI channels from Base, spreading a polyphonic part across a multitimbral rig - one note per channel, the classic voice-allocation split (distinct from MPEConvert's per-note MPE channels).

| Param | Range | Default | Unit |
|---|---|---|---|
| Voices | 1 – 16 | 4 | – |
| Base | 1 – 16 | 1 | – |

## Thin 1 input

Note-density limiter: drops any note-on arriving within MinGap ms of the last one passed, decluttering a busy stream or fast repeats - distinct from NoteLimit, which caps simultaneous polyphony. A dropped note's off is dropped too.

| Param | Range | Default | Unit |
|---|---|---|---|
| MinGap | 1 – 500 | 30 | ms |

# Sequencing

39 modules

## Arp 1 input

Arpeggiator: cycles held notes through a pattern at a synced rate with gate, octaves, swing and step count. Latch holds the chord so it keeps arpeggiating after the keys are released (the next key starts a new chord).

| Param | Range | Default | Unit |
|---|---|---|---|
| Mode | Up · Down · Up/Down · Down/Up · Converge · Diverge · As Played · Random · Chord | | – |
| Rate | 1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T | | – |
| Gate | 0 – 2 | 0.5 | – |
| Octaves | 1 – 4 | 1 | – |
| Swing | 0 – 0.75 | 0 | – |
| Steps | 1 – 16 | 16 | – |
| Latch | 0 – 1 | 0 | – |

## SeqEuclid 1 input

Euclidean rhythm gate: spreads Pulses evenly across Steps (rotatable) at a synced rate.

| Param | Range | Default | Unit |
|---|---|---|---|
| Pulses | 1 – 16 | 5 | – |
| Steps | 1 – 16 | 8 | – |
| Rotate | 0 – 15 | 0 | – |
| Rate | 1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T | | – |
| Gate | 0 – 2 | 0.5 | – |

## MidiGameOfLife 1 input

Conway's Game of Life MIDI sequencer: an 8x8 cell grid evolves under the B3/S23 rule; each clock step reads the next column and plays a note for every live cell (row maps to a scale degree from Root/Scale), and a full 8-step sweep advances one generation. Gliders, blinkers and still-lifes become ever-shifting melodic and harmonic patterns. Density sets the random fill when the grid (re)seeds. The MIDI-note counterpart to the GameOfLife CV block.

| Param | Range | Default |
|---|---|---|
| Root | C · C# · D · D# · E · F · F# · G · G# · A · A# · B | |
| Scale | Chromatic · Major · Natural Minor · Harmonic Minor · Melodic Minor · Dorian · Phrygian · Lydian · Mixolydian · Locrian · P | |
| Rate | 1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T | |
| Gate | 0 – 2 | 0.5 |
| Density | 0 – 1 | 0.35 |

## MidiBrianBrain 1 input

Brian's Brain MIDI sequencer: an 8x8 three-state grid (ready / firing / dying) evolves under Brian's Brain rules - a ready cell ignites only with exactly two firing neighbours, firing cells pass to dying, dying cells reset. Each clock step plays a note for every firing cell in the next column (row maps to a scale degree from Root/Scale); a full 8-step sweep advances one generation. The refractory state keeps it restless and sparkly where Game of Life settles. Density sets the random fill. The MIDI-note counterpart to the BrianBrain CV block.

| Param | Range | Default |
|---|---|---|
| Root | C · C# · D · D# · E · F · F# · G · G# · A · A# · B | |
| Scale | Chromatic · Major · Natural Minor · Harmonic Minor · Melodic Minor · Dorian · Phrygian · Lydian · Mixolydian · Locrian · P | |
| Rate | 1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T | |
| Gate | 0 – 2 | 0.5 |
| Density | 0 – 1 | 0.3 |

## MidiLangton 1 input

Langton's Ant MIDI melody: a single turmite walks an 8x8 grid - turning right on a white cell, left on a black one, flipping each cell as it leaves, then stepping forward. Each clock advances the ant Steps cells and plays one note for its current row (mapped to a scale degree from Root/Scale), so the melody scribbles chaotically then locks into the famous periodic 'highway'. Emergent order from one trivial rule - a single moving agent, not a population.

| Param | Range | Default |
|---|---|---|
| Root | C · C# · D · D# · E · F · F# · G · G# · A · A# · B | |
| Scale | Chromatic · Major · Natural Minor · Harmonic Minor · Melodic Minor · Dorian · Phrygian · Lydian · Mixolydian · Locrian · Pen | |
| Rate | 1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T | |
| Gate | 0 – 2 | 0.5 |
| Steps | 1 – 16 | 1 |

## Swing 1 input

Pushes off-beat eighth-notes later for a swing/shuffle feel.

| Param | Range | Default | Unit |
|---|---|---|---|
| Amount | 0 – 0.75 | 0.3 | – |

## PatternGate 1 input

Gates note-ons through a 16-step on/off pattern (bitmask).

| Param | Range | Default | Unit |
|---|---|---|---|
| Pattern | 0 – 65535 | 21845 | – |
| Steps | 1 – 16 | 16 | – |

## NoteOffDelay 1 input

Extends each note's gate by delaying its note-off.

| Param | Range | Default | Unit |
|---|---|---|---|
| Delay | 0 – 2000 | 200 | ms |

## Ricochet 1 input

Accelerating bouncing retriggers with decaying velocity.

| Param | Range | Default | Unit |
|---|---|---|---|
| Bounces | 1 – 16 | 6 | – |
| Decay | 0.3 – 0.98 | 0.7 | – |
| Start | 10 – 500 | 100 | ms |

## StepSeqMidi 1 input

Melodic step sequencer - semitone offsets (node config) clock the held root.

| Param | Range | Default | Unit |
|---|---|---|---|
| Rate | 1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T | | – |
| Gate | 0.05 – 1 | 0.5 | – |
| Velocity | 1 – 127 | 100 | – |

## Echo 1 input

MIDI delay: repeats notes at a synced time with velocity feedback decay.

| Param | Range | Default | Unit |
|---|---|---|---|
| Time | 1 – 2000 | 250 | ms |
| Feedback | 0 – 0.95 | 0.5 | – |
| Repeats | 1 – 8 | 3 | – |

## Flam 1 input

Precedes every note with a quiet grace note, delaying the main hit so the grace leads it - the classic drum flam.

| Param | Range | Default | Unit |
|---|---|---|---|
| Time | 1 – 200 | 30 | ms |
| Grace Vel | 0 – 1 | 0.5 | – |

## Roll 1 input

Drum roll / buzz: retriggers every held note at Rate for as long as it is held, each hit gated to a fraction of the interval.

| Param | Range | Default | Unit |
|---|---|---|---|
| Rate | 1 – 50 | 12 | Hz |
| Gate | 0.05 – 0.95 | 0.5 | – |

## Trill 1 input

Keyboard trill: while a note is held, alternates rapidly between it and the note Interval semitones above, at Rate.

| Param | Range | Default | Unit |
|---|---|---|---|
| Rate | 1 – 50 | 10 | Hz |
| Interval | 1 – 12 | 2 | st |
| Gate | 0.05 – 0.95 | 0.5 | – |

## Mordent 1 input

One-shot ornament: each struck note plays principal -> auxiliary -> principal at the attack, then sustains, the single-flick counterpart to the continuous Trill.

| Param | Range | Default | Unit |
|---|---|---|---|
| Time | 5 – 150 | 40 | ms |
| Interval | 1 – 12 | 2 | st |

| Param | Range | Default | Unit |
|---|---|---|---|
| Direction | Upper · Lower | | – |

## Turn 1 input

Gruppetto turn: a four-step ornament at the attack - upper auxiliary, principal, lower auxiliary, then the principal held. Time sets each step, Interval the neighbour distance.

| Param | Range | Default | Unit |
|---|---|---|---|
| Time | 5 – 150 | 35 | ms |
| Interval | 1 – 12 | 2 | st |

## Appoggiatura 1 input

Leaning grace note: each struck note first sounds a long auxiliary (Interval away) that takes Time from the principal, which then sustains - the expressive long grace, unlike the quick Flam or Mordent.

| Param | Range | Default | Unit |
|---|---|---|---|
| Time | 20 – 400 | 120 | ms |
| Interval | 1 – 12 | 2 | st |
| Direction | Upper · Lower | | – |

## Strum 1 input

Spreads a chord's notes over time, up or down, like a guitar strum.

| Param | Range | Default | Unit |
|---|---|---|---|
| Spread | 0 – 200 | 20 | ms |
| Direction | Up · Down | | – |

## MidiQuantize 1 input

Snaps note starts to a synced grid by a strength amount.

| Param | Range | Default | Unit |
|---|---|---|---|
| Rate | 1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T | | – |
| Strength | 0 – 1 | 1 | – |

## SeqRatchet 1 input

Retriggers each note Count times within a synced step.

| Param | Range | Default | Unit |
|---|---|---|---|
| Rate | 1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T | | – |
| Count | 1 – 8 | 4 | – |
| Gate | 0.05 – 1 | 0.9 | – |

## NoteLength 1 input

Forces a fixed note duration (gate length).

| Param | Range | Default | Unit |
|---|---|---|---|
| Length | 1 – 2000 | 200 | ms |

## Cascade 1 input

Each struck note fans out into a timed run of Steps notes climbing (or falling) the scale from it - an arpeggiated flourish fired per key, unlike Arp which cycles the whole held chord.

| Param | Range | Default |
|---|---|---|
| Root | C · C# · D · D# · E · F · F# · G · G# · A · A# · B | |
| Scale | Chromatic · Major · Natural Minor · Harmonic Minor · Melodic Minor · Dorian · Phrygian · Lydian · Mixolydian · Locrian · | |
| Steps | 2 – 8 | 4 |
| Time | 5 – 200 | 50 |
| Direction | Up · Down | |

## MidiTuring 1 input

Turing-machine sequencer (Music Thing style): a circular shift register clocks once per step and the held notes fire when the read bit is 1. Change is the probability the recycled bit flips - 0 locks an evolving loop, 1 fully randomises, between = slowly mutating melodies.

| Param | Range | Default | Unit |
|---|---|---|---|
| Rate | 1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T | | – |
| Length | 2 – 16 | 8 | – |
| Change | 0 – 1 | 0.15 | – |
| Gate | 0.05 – 1 | 0.5 | – |

## Polymeter 1 input

Two step lanes of different lengths (LenA, LenB) clock together over the held notes, phasing against each other so the combined pattern only repeats after lcm(LenA,LenB) steps - instant polymetric movement from one chord.

| Param | Range | Default | Unit |
|---|---|---|---|
| Rate | 1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T | | – |
| Len A | 1 – 16 | 3 | – |
| Len B | 1 – 16 | 4 | – |
| Gate | 0.05 – 1 | 0.5 | – |

## MidiTranceGate 1 input

Chops the held notes with a 16-step on/off Pattern at a synced Rate: every 'on' step retriggers all held notes, gated to a fraction of the step. The rhythmic gate behind the trance sound - pattern-driven, where Roll is a fixed pulse.

| Param | Range | Default | Unit |
|---|---|---|---|
| Rate | 1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T | | – |
| Pattern | 0 – 65535 | 21845 | – |
| Steps | 1 – 16 | 16 | – |
| Gate | 0.05 – 1 | 0.5 | – |

## StepArp 1 input

Cthulhu-style step arpeggiator: walks the held chord strictly upward across Octaves octaves at a synced Rate, accenting the first step of each cycle - the rolling, octave-jumping arp of Xfer Cthulhu's sequencer. Distinct from Arp's mode-cycling: this climbs the chord in order.

| Param | Range | Default | Unit |
|---|---|---|---|
| Rate | 1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T | | – |
| Gate | 0.05 – 1 | 0.5 | – |
| Octaves | 1 – 4 | 2 | – |
| Accent | 1 – 127 | 110 | – |

## ChordArp 1 input

Cthulhu-style chord arpeggiator: each key builds a full chord (Type) and the result is arpeggiated upward across Octaves at a synced Rate - one finger plays a moving arpeggio of the whole chord. Releasing the key stops its run.

| Param | Range | Default |
|---|---|---|
| Type | Unison · Octave · 5th (Power) · 5th + Oct · Tritone · Major · Minor · Diminished · Augmented · Major (1st inv) · Major (2n | |
| Rate | 1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T | |
| Gate | 0.05 – 1 | 0.5 |
| Octaves | 1 – 4 | 1 |

## OctaveArp 1 input

Cthulhu's arp octave lane: arpeggiates the held chord while a per-step octave-offset Pattern leaps the line up and down by whole octaves (alternating, staircase, bounce...) instead of climbing monotonically like StepArp.

| Param | Range | Default | Unit |
|---|---|---|---|
| Pattern | 0 – 7 | 1 | – |
| Rate | 1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T | | – |
| Gate | 0.05 – 1 | 0.5 | – |

## GateArp 1 input

Cthulhu's arp gate lane: walks the held chord upward across Octaves while the note length cycles a per-step staccato/legato pattern, so the rhythm breathes - short stabs then sustained notes. Gate scales the pattern (>1 overlaps into the next step).

| Param | Range | Default | Unit |
|---|---|---|---|
| Rate | 1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T | | – |
| Gate | 0.25 – 2 | 1 | – |
| Octaves | 1 – 4 | 1 | – |

## VelArp 1 input

Cthulhu's arp velocity lane as a ramp: arpeggiates the held chord while velocity sweeps from From to To across Steps then resets, so every pass swells or fades without a per-step editor.

| Param | Range | Default | Unit |
|---|---|---|---|
| Rate | 1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T | | – |
| Gate | 0.05 – 1 | 0.5 | – |
| From | 1 – 127 | 40 | – |

| Param | Range | Default | Unit |
| --- | --- | --- | --- |
| To | 1 – 127 | 120 | – |
| Steps | 1 – 32 | 8 | – |

## TransArp 1 input

Cthulhu's arp transpose lane: arpeggiates the held chord upward across Octaves while a per-step semitone Pattern (fifth pedals, triad arps, scalar runs) is added on top, so the line outlines a melodic contour, not just the chord tones.

| Param | Range | Default | Unit |
| --- | --- | --- | --- |
| Pattern | 0 – 7 | 2 | – |
| Rate | 1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T | | – |
| Gate | 0.05 – 1 | 0.5 | – |
| Octaves | 1 – 4 | 1 | – |

## RatchetArp 1 input

Cthulhu's arp repeat lane: walks the held chord upward across Octaves while a per-step pattern retriggers selected steps multiple times (capped by Max), packing rolls into the line. Unlike SeqRatchet's fixed count, the repeat count varies per step.

| Param | Range | Default | Unit |
| --- | --- | --- | --- |
| Rate | 1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T | | – |
| Gate | 0.05 – 1 | 0.5 | – |
| Octaves | 1 – 4 | 1 | – |
| Max | 1 – 8 | 4 | – |

## SkipArp 1 input

Cthulhu's arp skip lane: arpeggiates the held chord upward across Octaves, but each step has a Skip probability of being dropped to a rest, thinning the pattern differently every cycle for generative movement.

| Param | Range | Default | Unit |
| --- | --- | --- | --- |
| Rate | 1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T | | – |
| Gate | 0.05 – 1 | 0.5 | – |
| Octaves | 1 – 4 | 1 | – |
| Skip | 0 – 0.95 | 0.3 | – |

## Drunk 1 input

Random-walk melodic sequencer: each synced Rate step nudges the pitch by a random +/-Step semitones, bounded within +/-Range of the lowest held note - Brownian melody anchored to the chord, unlike MidiRandom's memoryless picks.

| Param | Range | Default | Unit |
| --- | --- | --- | --- |
| Rate | 1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T | | – |
| Gate | 0.05 – 1 | 0.5 | – |
| Step | 1 – 7 | 2 | st |
| Range | 1 – 24 | 12 | st |

## EuclidMel 1 input

Euclidean melody: spreads Pulses evenly across Steps (Bjorklund) and, on each hit, advances a cursor through the held chord - the rhythm is Euclidean and the pitch walks the chord, unlike SeqEuclid which stabs all held notes per pulse.

| Param | Range | Default | Unit |
|---|---|---|---|
| Pulses | 1 – 16 | 5 | – |
| Steps | 1 – 16 | 8 | – |
| Rate | 1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T | | – |
| Gate | 0.05 – 1 | 0.5 | – |

## Spiral 1 input

Transposing MIDI delay: each repeat is delayed by Time and shifted by Interval semitones with velocity decaying by Feedback, so one note spins off a climbing or falling staircase of echoes - unlike Echo, which repeats the same pitch.

| Param | Range | Default | Unit |
|---|---|---|---|
| Time | 1 – 1000 | 120 | ms |
| Interval | -12 – 12 | 4 | st |
| Feedback | 0 – 0.95 | 0.7 | – |
| Repeats | 1 – 12 | 5 | – |

## Tuplet 1 input

Snaps note starts to a Tuplets-per-beat grid (3 = triplets, 5 = quintuplets) by Strength, for a triplet or odd-tuplet feel where MidiQuantize only snaps to the straight grid.

| Param | Range | Default | Unit |
|---|---|---|---|
| Tuplets | 2 – 9 | 3 | – |
| Strength | 0 – 1 | 1 | – |

## Shift 1 input

Delays every note by Delay ms (groove nudge / lay-back), pushing note-ons and offs back by the same amount so durations are preserved - the per-track timing offset of a groove sequencer.

| Param | Range | Default | Unit |
|---|---|---|---|
| Delay | 0 – 200 | 20 | ms |

## MidiMarkov 1 input

Learns the note-to-note transition statistics of what you play (a 12x12 pitch-class chain) and, on each synced step, walks the chain to improvise a new line in the same style. Memory fades old transitions.

| Param | Range | Default | Unit |
|---|---|---|---|
| Rate | 1/1 · 1/2 · 1/4 · 1/8 · 1/16 · 1/32 · 1/4. · 1/8. · 1/16. · 1/4T · 1/8T · 1/16T | | – |
| Gate | 0.05 – 1 | 0.5 | – |
| Memory | 0 – 1 | 0.2 | – |

# Tuning

3 modules

## Microtuner 1 input

Stretched-octave microtuning via per-note pitch detune.

| Param | Range | Default | Unit |
|---|---|---|---|
| Stretch | -50 – 50 | 0 | cents/oct |

## JustTune 1 input

Adaptive just intonation: retunes each note toward its 5-limit JI ratio relative to Key via per-note detune (polyphonic), Amount blending from equal temperament to pure - the beatless thirds and fifths of real harmony.

| Param | Range | Default | Unit |
|---|---|---|---|
| Key | C · C# · D · D# · E · F · F# · G · G# · A · A# · B | | — |
| Amount | 0 – 1 | 1 | — |

## TuneTable 1 input

Custom microtuning: detunes each note by a per-pitch-class cents offset from the node's 12-value config list (relative to Key) - meantone, Pythagorean, maqam or gamelan, any temperament. Amount scales the table.

| Param | Range | Default | Unit |
|---|---|---|---|
| Key | C · C# · D · D# · E · F · F# · G · G# · A · A# · B | | — |
| Amount | 0 – 1 | 1 | — |

# Voicing

5 modules

## MPEConvert 1 input

Assigns each note its own rotating channel for per-note MPE expression.

| Param | Range | Default | Unit |
|---|---|---|---|
| Voices | 2 – 15 | 15 | – |

## Sustain 1 input

Sustain-pedal simulation: while Hold is engaged, note-offs are captured so the notes keep sounding; when Hold drops, every held note releases at once. Hold is a parameter, so a CC, an LFO or automation can play the pedal.

| Param | Range | Default | Unit |
|---|---|---|---|
| Hold | Off · On | | – |

## Mono 1 input

Monophonic note priority (last / lowest / highest).

| Param | Range | Default | Unit |
|---|---|---|---|
| Priority | Last · Lowest · Highest | | – |

## MidiUnison 1 input

Doubles each note into multiple voices, optionally spread across channels.

| Param | Range | Default | Unit |
|---|---|---|---|
| Voices | 1 – 4 | 2 | – |
| Spread | Same chan · Spread chans | | – |

## Slide 1 input

Cthulhu's slide: mono legato glue. Picks one held note by Priority (last/low/high) and emits the new note-on before releasing the old one, so a portamento synth slurs between notes instead of retriggering.

| Param | Range | Default | Unit |
|---|---|---|---|
| Priority | Last · Low · High | | – |